

NITISH HARSOOR

Data Scientist

9663956073 • nharsoor048@gmail.com • <https://www.linkedin.com/in/nitish-harsoor/> • <https://nitishkmr005.github.io/> • Bangalore

Summary

Data Scientist specializing in **Generative AI, LLM evaluation, recommender systems, and NLP**. Proven expertise in building **production-grade ML and LLM systems** end-to-end—from problem framing and model development to deployment and monitoring. Delivered **\$197M+ business impact**, enabled **64,600+ incremental user engagements**, and shipped multiple **ARB-approved enterprise model deployments** in large-scale financial services environments.

Skills and Certifications

♦ Generative AI & LLMs:

Large Language Models (LLM), LLM as a Judge, LLM Evaluation, LLM Leaderboard, RAG, Prompt Engineering, Chain Of Thought Prompting, Reflection Agents, Information Extraction, Voice Agents, Speech to Text (STT), Text to Speech (TTS), T5 Fine-tuning, LangChain, LangGraph, FAISS, Vector Databases, Transformers

♦ Machine Learning & NLP:

Machine Learning, Recommender Systems, Two-Tower, LightGBM, XGBoost, Neural Networks, BERT, Feature Engineering, Model Explainability (SHAP), Fairness and Bias Analysis (Jurity), Scikit-Learn, Pandas, Feature Selection, Sentence Transformers, Precision/Recall/F1/CTR Metrics, Lift Report Creation, A/B Testing

♦ Engineering & MLOps:

Python, PyTorch, Streamlit, FastAPI, Pydantic, Docker, Linux, AWS SageMaker, Bedrock, S3, Snowflake, SQL, SQL Joins, SQL CTE, SQL Window Functions, Tableau

♦ Coursera Certifications:

 Natural Language Processing Specialization, Deep Learning Specialization, Machine Learning by Andrew Ng

Experience

Fidelity Investments

Bangalore, India

Data Scientist

11/2022 - Present

Fidelity Investments is a global financial services leader managing \$5.8T in assets, offering brokerage, mutual funds, retirement services, and wealth management solutions.

- Developed the **Workplace Seminar Recommendation System** for over 15 million participants using two stage architecture utilizing LightGBM and neural network, delivering a **2.3× lift in event attendance, adding 64,600+ participants**, and generating over **\$197 Million in incremental P/W flows**.
- Led AgentBot 3.0, building an enterprise GenAI pipeline where LLM-extracted structured JSON from call transcripts powered downstream pre-call insights, in-call assistance, and post-call rep note/email generation—**driving \$32M annual cost savings, 44% reduction in after-call work, and 1% increased appointment rate**.
- Deployed the Helios Collections recommender to production using LinGreedy contextual multi-armed bandits, modeling each collection as an arm and learning context-aware rewards via ridge regression, resulting in a **2.13× lift in CTR@1** over the baseline in the Helios mobile app.
- Fine-tuned T5 transformer models for **email summarization** on noisy email chains, combining regex-based parsing and embedding-driven noise filtering to isolate salient content and improve summary quality.
- Implemented **model interpretability and fairness evaluation** for every recommender systems using **SHAP and Jurity libraries**, improving transparency, reliability, and bias detection.
- Built an end-to-end **voice agent pipeline**—covering **speech-to-text, turn detection, LLM-based tool calling, and text-to-speech**—as part of a Fidelity GenAI hackathon initiative.

Fidelity Investments

Bangalore, India

Lead Systems Analyst

04/2021 - 10/2022

- Built a retrieval-and-re-ranker-based neural search engine for the Gurubar tech support app, enabling associates to self-resolve laptop issues and **reducing ticket volume by ~40%** through relevant KB article recommendations.
- Built enterprise-grade ML systems including a holiday incident classifier and BERT-based clustering engine, automating manual ticket review and achieving **90%+ accuracy** in identifying repeated incidents during peak periods and enabling proactive prevention of recurring incidents.
- Developed ML-driven ticket routing models (XGBoost + BERT NLP) that achieved **85%+ assignment accuracy**, reducing MTTR by routing tickets to correct teams automatically.
- Architected and deployed to production Data Intelligence Platform serving 720+ users who can view all dashboards in one place, implementing Dockerized infrastructure, NGINX load handling concurrent connections and enterprise-grade SSO/SSL security.

Experience

Fidelity Investments

Bangalore, India

Software Engineer

04/2016 - 04/2021

- Worked as a Software Engineer on the Purchase and Sales (P&S) application, a trade processing system that captures trade data from multiple feeds such as order-entry systems, validates and processes trades, and delivers formatted transactions to back-office systems via a dispatcher component. The system also manages trade exceptions, supports reprocessing of failed transactions, and provides functionality to cancel or correct trades.

IBM

Bangalore, India

System Engineer

04/2013 - 04/2016

IBM (International Business Machines Corporation) is a leading American multinational technology company that provides hybrid cloud and AI solutions, software, services, and mainframe computers.

- Worked in Universal Payment System (UPS) application, supporting large-scale cheque issuance, ACH payments, and generation of EOP/EOB statements for healthcare providers and members.

Projects

AgentBot 3.0 - Streamline Contact Centre Associate Task using GenAI Capabilities

10/2025

Streamline contact centre associate task and routines by automating experiences at different stages of the call (pre-call, during, after) using state-of-the-art LLM Models.

- Designed a two-stage GenAI pipeline where **structured JSON extraction** served as a data foundation for downstream rep note, email, and pre-call insight generation.
- Built an **LLM-as-a-Judge evaluation framework** with attribute-level precision/recall, missing vs incorrect signals, and model leaderboards for systematic comparison.
- Built a **production-grade LLM validation platform** using streamlit for validating **150+ extracted attributes across 2,000+ calls**, enabling hallucination detection, missing-field analysis, real-time dashboards, and automated prompt optimization—replacing months of manual QA.
- Leveraged **Claude 3.5 Sonnet** to extract **150+ structured attributes** from call transcripts, enforcing output correctness through **Pydantic-based JSON schema validation**.

Contextual Multi-Armed Bandit Recommender for Investment Collections

09/2024

The goal of this project was to personalize the **Collections** component in the Helios mobile app to increase customer engagement, where each collection represents an **investment idea** composed of thematically similar underlying products.

- Implemented a **LinGreedy contextual multi-armed bandit**, modeling each investment collection as an arm with **per-arm ridge regression**.
- We utilized **Mab2Rec** library to evaluate linear policies from MABWiser and benchmarked these against non-contextual random policies.

Workplace Seminar Recommendation

11/2023

The objective was to identify the **most relevant seminar topics for each participant** and aggregate these recommendations at the **plan-sponsor level**, enabling targeted and personalized communication campaigns.

- Designed two stage modeling approach - Stage1 predict the likelihood of a participant to attend any seminar and Stage2 predict the likelihood of a participant to attend a particular theme.
- Implemented a **LightGBM binary classifier** for Stage 1 and a **neural network-based multi-label classifier** for Stage 2, trained on **two years of historical attendance data** and validated on **out-of-time datasets** with consistent performance.

Awards

💎 EIA Award 2025 - AgentBot 3.0 development and GenAI Capabilities

💎 Eureka Award H2 2024 - Charter Champion category for Workplace Seminar Recommendation

💎 EIA Award 2024 - AI model migration from CWE to EDL (WPS & Helios Collections)

💎 Excellent Performer 2025 - Year End Performance Rating

Education

PES Institute of Technology - Bangalore

08/2010 - 10/2012

M. Tech in Digital Electronics